INSA INSTITUT NATIONAL DES SCIENCES APPLIQUÉES HAUTS-DE-FRANCE

Université Polytechnique HAUTS-DE-FRANCE

iut. INSTITUT UNIVERSITAIRE DE TECHNOLOGIE VALENCIENNES • CAMBRAI • MAUBEUGE
Département Génie électrique et informatique industrielle
Université Polytechnique HAUTS-DE-FRANCE

ARPEX

# End-of-project review

PLP25INT26 - IUT GEII & INSA Hauts-de-France

Supervisors:
Nicolas Lardenois
Marc Rosique



# Contents

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V. Aerodynamics study
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# Overview: Electric Concept Car

**Our goal is unchanged: to develop an electric racing concept car with indirect vision through cameras, pushing the boundaries of automotive innovation.**



## Bodywork & Chassis






## Aerodynamics






## Electronics & Wheel

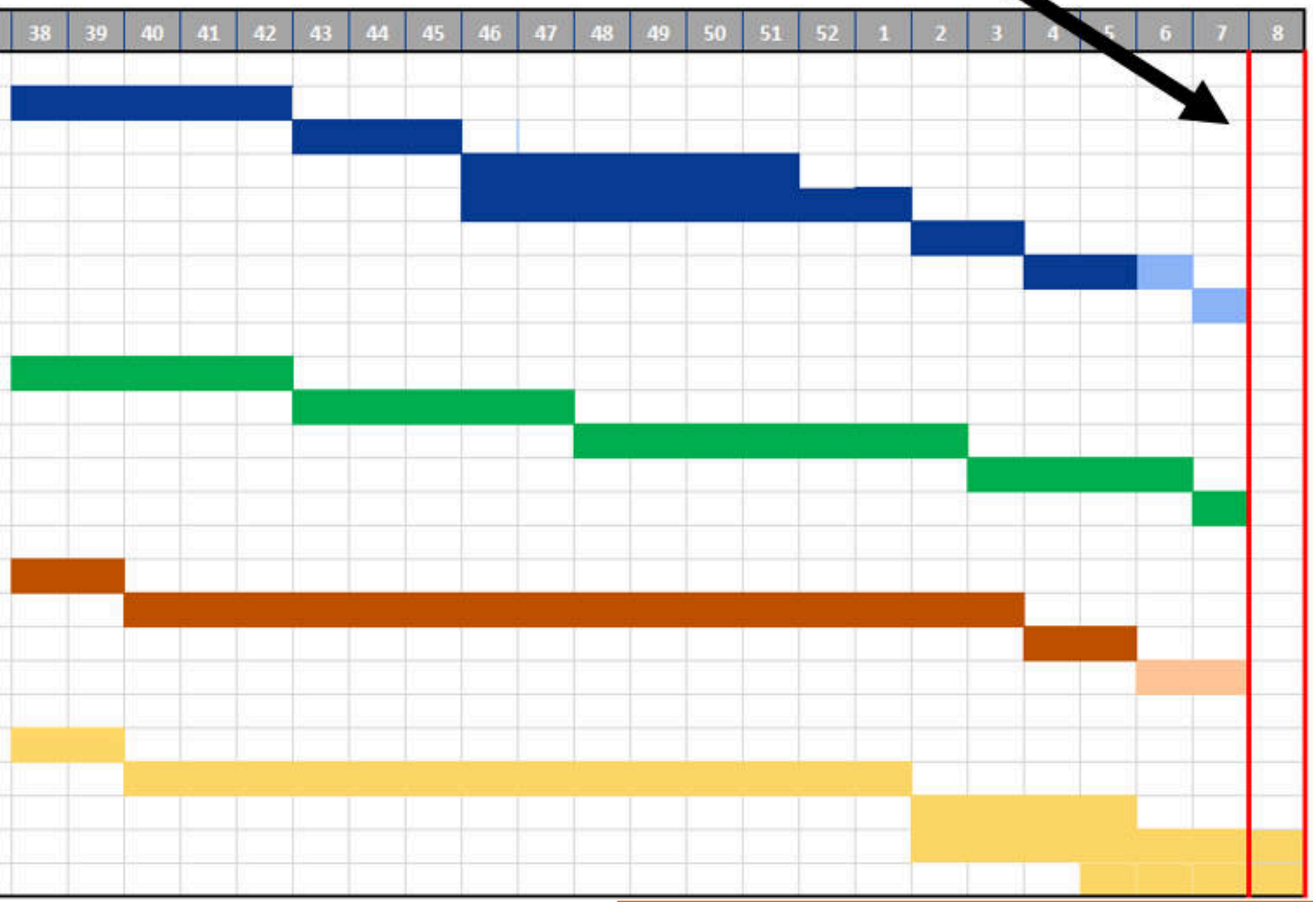


# Progress chart

| | |
|---|---|
| Nom du projet : | ARPEX |
| Début du projet : | mercredi,17-septembre-2025 |
| Fin du projet : | mardi,17-février-2026 |

Cette semaine

| Tâches | Date de Début | Date de Fin | Statut | Progression |
|---|---|---|---|---|
| **Carosserie & Aérodynamique** | | | | |
| Recherches bibliographiques | 17/09/25 | 20/10/25 | Terminée | 100% |
| Choix des modèles | 20/10/25 | 10/11/25 | Terminée | 100% |
| Assemblage du model | 10/11/25 | 22/12/25 | Terminée | 100% |
| Préparation du modèle physique sur logiciel | 10/11/25 | 5/01/26 | Terminée | 100% |
| Création du maillage | 5/01/26 | 19/01/26 | Terminée | 100% |
| Simulations numériques et ajustements | 19/01/26 | 9/02/26 | En cours | 90% |
| Analyse des Résultats | 9/02/26 | 11/02/26 | En cours | 50% |
| **Conception Volant** | | | | |
| Analyse du sujet | 15/09/25 | 22/10/25 | Terminée | 100% |
| Choix des composants utilisés | 22/10/25 | 24/11/25 | Terminée | 100% |
| élaboration du Software | 24/11/25 | 14/01/26 | Terminée | 90% |
| élaboration du Hardware | 14/01/26 | 9/02/26 | Terminée | 100% |
| tests | 9/02/26 | 11/02/26 | Terminée | 100% |
| **Réalisation Chassis** | | | | |
| Analyse des travaux du groupe précédent | 15/09/25 | 24/09/25 | Terminée | 100% |
| Recherche d'optimisation du châssis | 24/10/25 | 12/01/26 | En cours | 100% |
| Mise en place du châssis final pour CFD | 12/01/26 | 26/01/26 | En cours | 95% |
| Tests structurels | 26/01/26 | 11/02/26 | En cours | 10% |
| **Management** | | | | |
| Plannification et définition du scope | 15/09/25 | 29/09/25 | Terminée | 100% |
| Suivi & gestion des risques | 29/09/25 | 5/01/26 | Terminée | 100% |
| Consolidation des livrables | 5/01/26 | 2/02/26 | En cours | 100% |
| Rédaction du rapport final | 5/01/26 | 16/02/26 | Non Commencé | 100% |
| Préparation soutenance finale | 9/02/26 | 16/02/26 | Non Commencé | 100% |



# PDCA

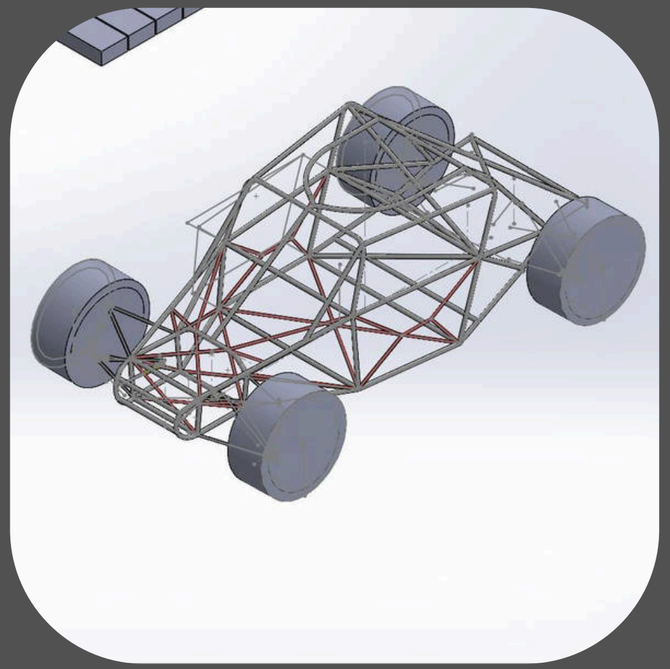
| P (Prévoir) | | D (Piloter) | | C (Vérifier) | | A (Pérenniser) | |
|---|---|---|---|---|---|---|---|
| Recherches Bibliographiques carosserie et aérodynamiques | Anthony [redacted] 17/09/25 V | Assemblage, Création du model de carrosserie | Anthony LAÏ-HONG-TING 19/12/25 V | Simulations numériques et ajustements | Anthony LAÏ-HONG-TING 06/02/26 O | Rédaction du rapport final et préparation de la soutenance | CP V |
| Choix des modèles pour la carosserie | Anthony LAÏ-HONG-TING 10/11/25 V | Préparation du modèle physique sur logiciel | Anthony LAÏ-HONG-TING 06/01/26 V | Performances aérodynamiques & validation des résultats CFD | Anthony LAÏ-HONG-TING 06/02/2026 O | Modification du modèle | Anthony LAÏ-HONG-TING O |
| Définition des livrables attendus : Modèles sélectionnés, documentation technique | Anthony LAÏ-HONG-TING 20/10/25 V | Création du maillage et simulation de la trainée et de l'appui | Anthony LAÏ-HONG-TING 14/01/26 V | Verification des Softwares | Martin BRETECHER 16/2/26 V | Amelioration continue des softwares | Martin BRETECHER P |
| Définition des objectifs aérodynamiques | Anthony LAÏ-HONG-TING 20/10/25 V | Suivi et gestion des risques | CP 02/02/26 V | Verification du Hardware | Martin BRETECHER 10/02/2026 V | Ajouter de nouvelles fonctions | Martin BRETECHER R |
| Analyse comparative des différentes géométries | Anthony LAÏ-HONG-TING 10/11/25 V | Création des softwares | Martin BRETECHER 12/11/25 V | Vérification de conformité avec les objectifs | CP 02/02/2026 O | Mise en place du châssis final et de ses composants | Lucas VILLEGAS 17/02/2026 V |
| Cahier des Charges complété et validé | CP 24/9/25 V | Création du modèle d'une boite de vitesse virtuelle | Martin BRETECHER 26/11/25 V | Consolidation des livrables intermédiaires | CP 12/01/2026 V | Permettre la meilleure continuité du projet | Lucas VILLEGAS 17/02/2026 V |
| Plannification et définition du scope | CP 20/10/25 V | Réalisation du Hardware | Martin BRETECHER 04/02/26 V | Tests de contraintes structurels | Lucas VILLEGAS R | | |
| Allocation des ressources | CP 20/10/25 V | Ajout des éléments supplémentaires sur le modèle | Lucas VILLEGAS 14/01/26 V | Vérifier les contraintes physiques | Lucas VILLEGAS R | | |
| Répertorier le matériel | Martin BRETECHER 22/10/25 V | Correction du modèle précédent | Lucas VILLEGAS 24/09/25 V | | | | |
| Modélisation de notre solution | Martin BRETECHER 29/9/25 V | Recherche d'optimisation du châssis | Lucas VILLEGAS V | | | | |
| Recherches bibliographiques | Martin BRETECHER 15/9/25 V | | | | | | |
| Recherches bibliographiques | Lucas VILLEGAS 17/9/25 V | | | | | | |
| Analyse des travaux du groupe précédent | Lucas VILLEGAS 24/9/25 V | | | | | | |
| Recherche de licence Solidworks | Lucas VILLEGAS 24/9/25 V | | | | | | |

| Référents | Domaines |
|---|---|
| CP | |
| Morgan SOUPAMA | Management |
| Lucas VILLEGAS | CAO + Chassis |
| Anthony LAÏ-HONG-TING | CFD / Aérodynamique |
| Martin BRETECHER | Mécatronique / Volant |

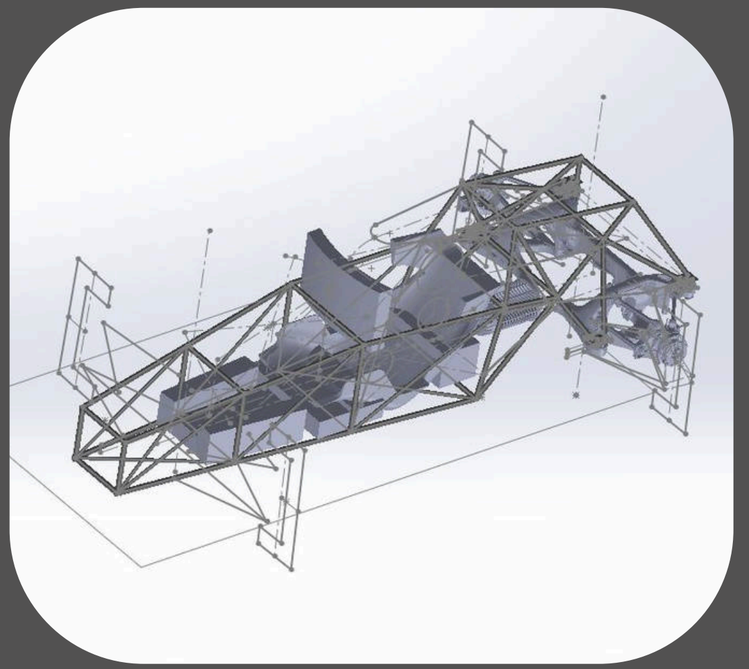
| | |
|---|---|
| P | Pas encore commencé |
| R | Tâche compromise |
| O | Résultat partiellement obtenu |
| V | Résultat obtenu |

# Chassis Modelling

## Initial chassis



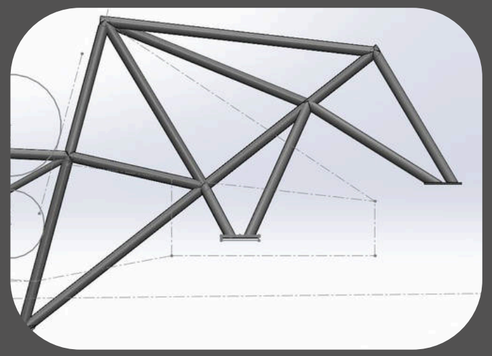
## Current chassis

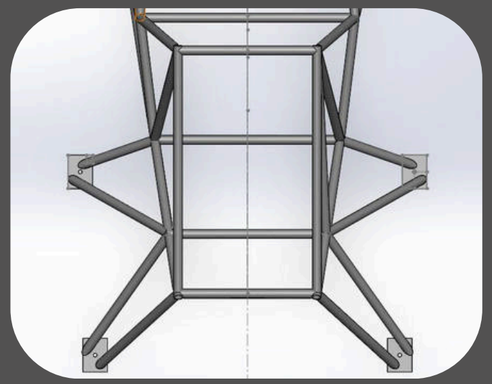


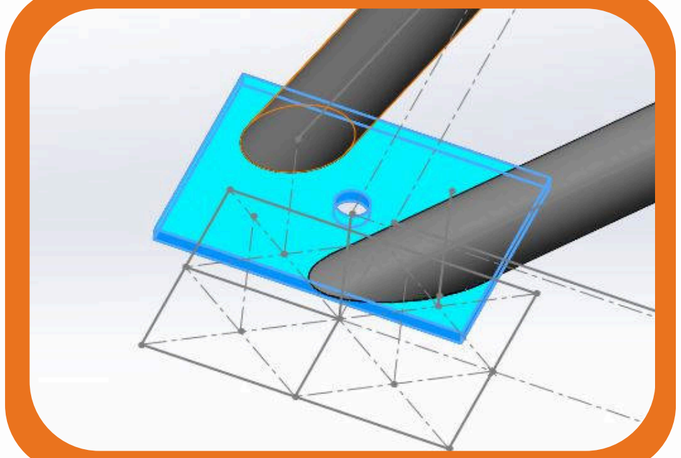
# Building the Chassis

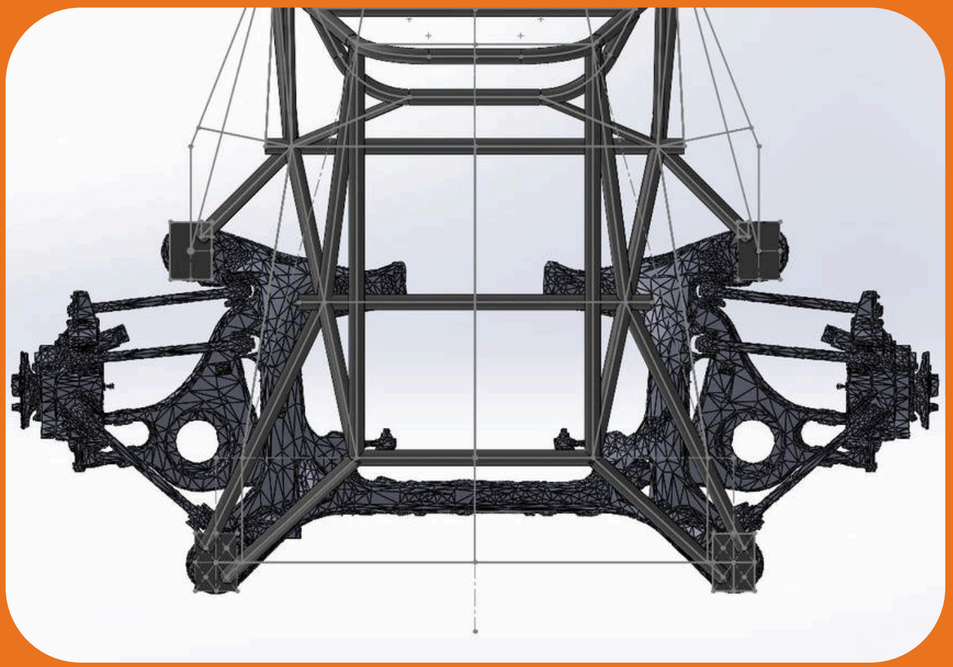
## Fitting the rear axle

- Placing the mounting points
- Correct sizing relative to the ground
- Making mounting plates

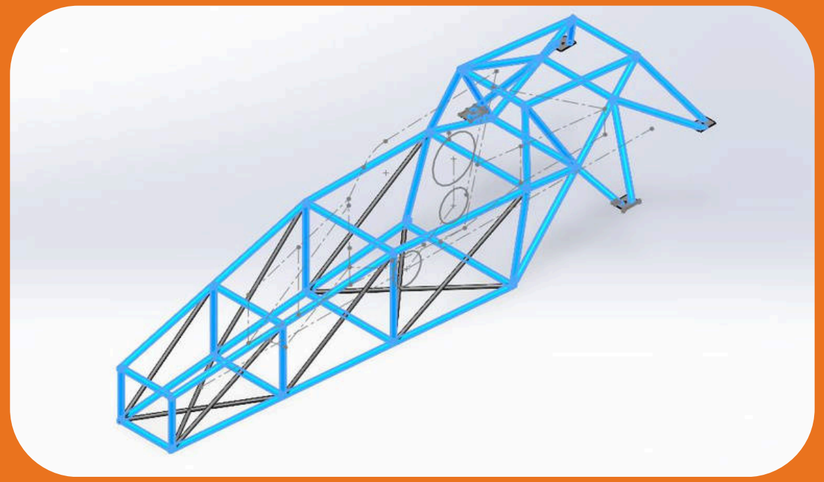


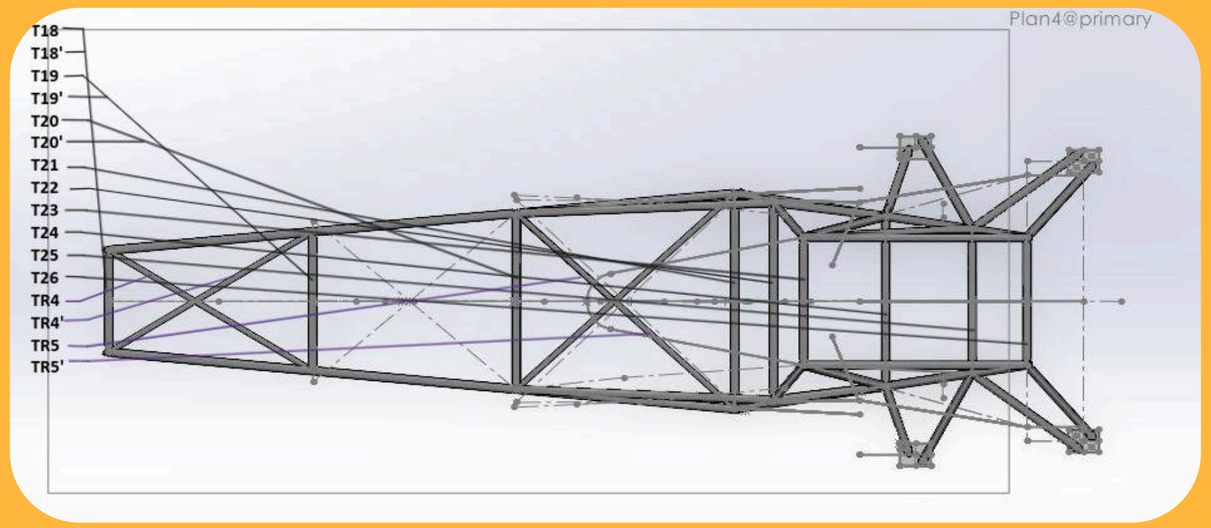


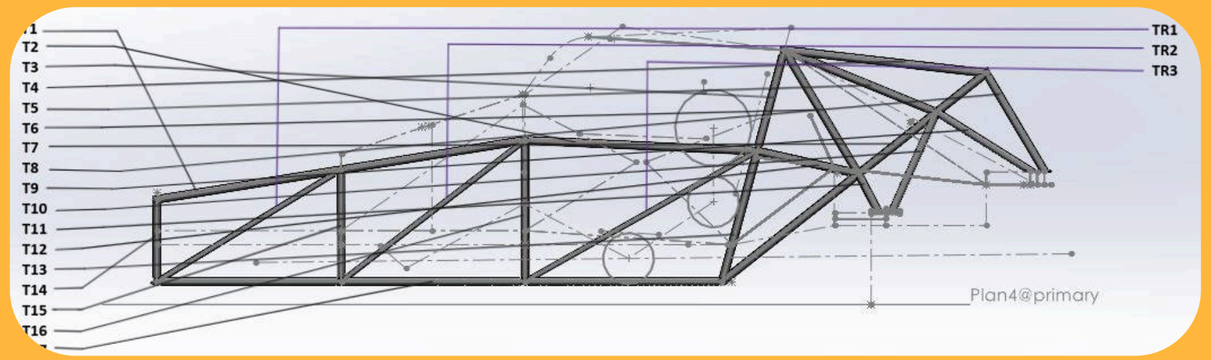




# Chassis - Tubing

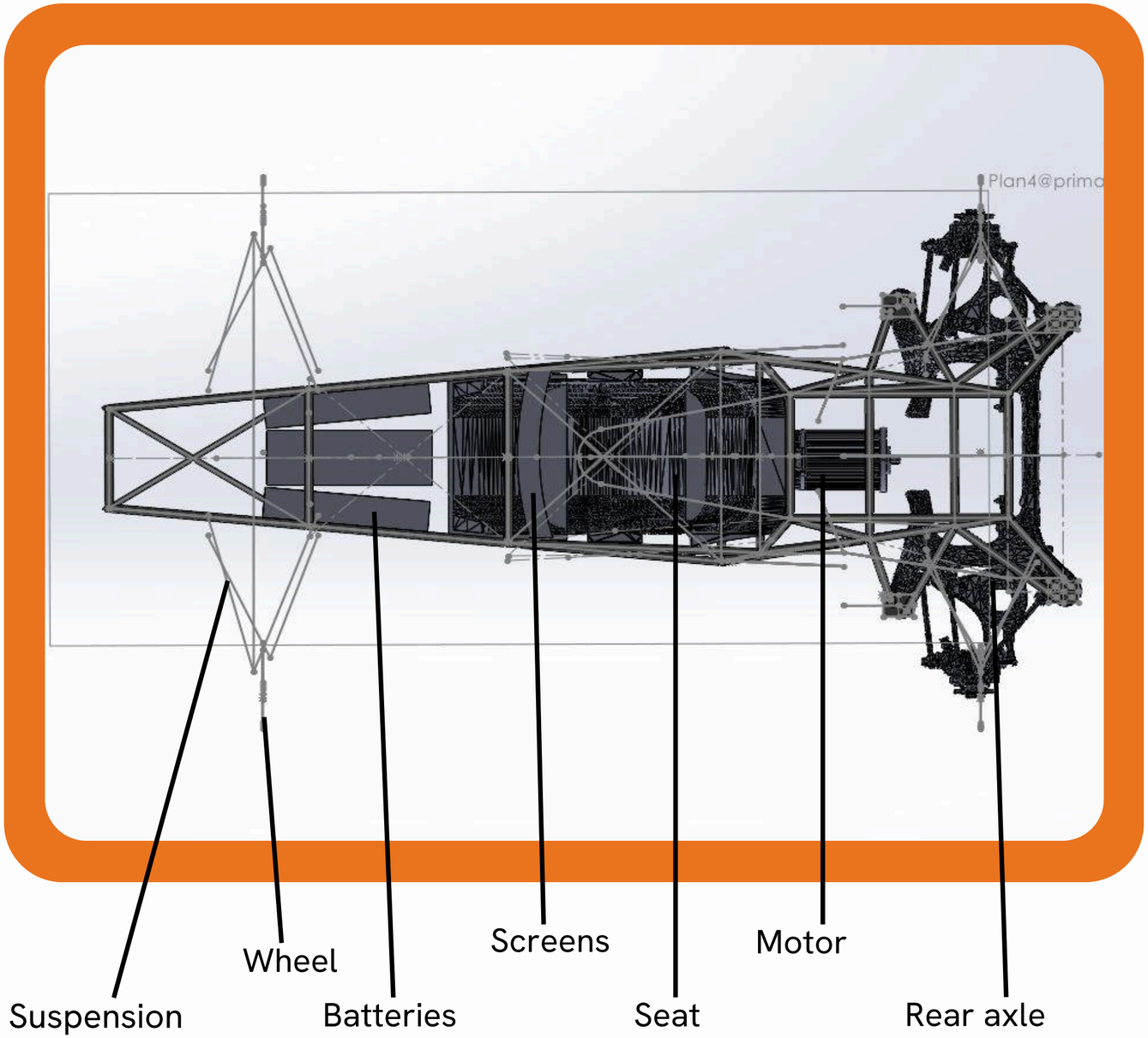




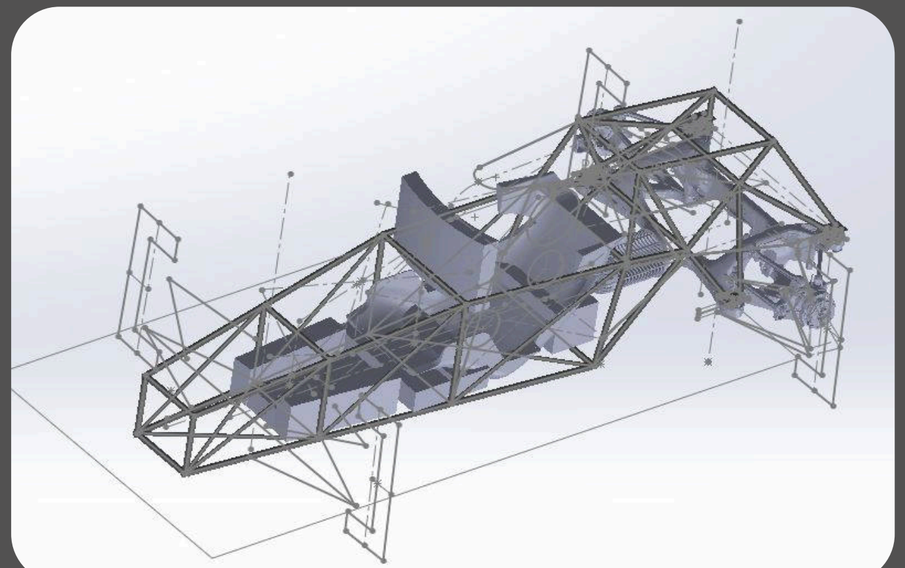




- 56 Tubes
- 40.8 metres of tube
  31.4 metres at 50 mm
  9.4 metres at 40 mm
- Reinforcement completed
- €1,100 of tubing incl. VAT

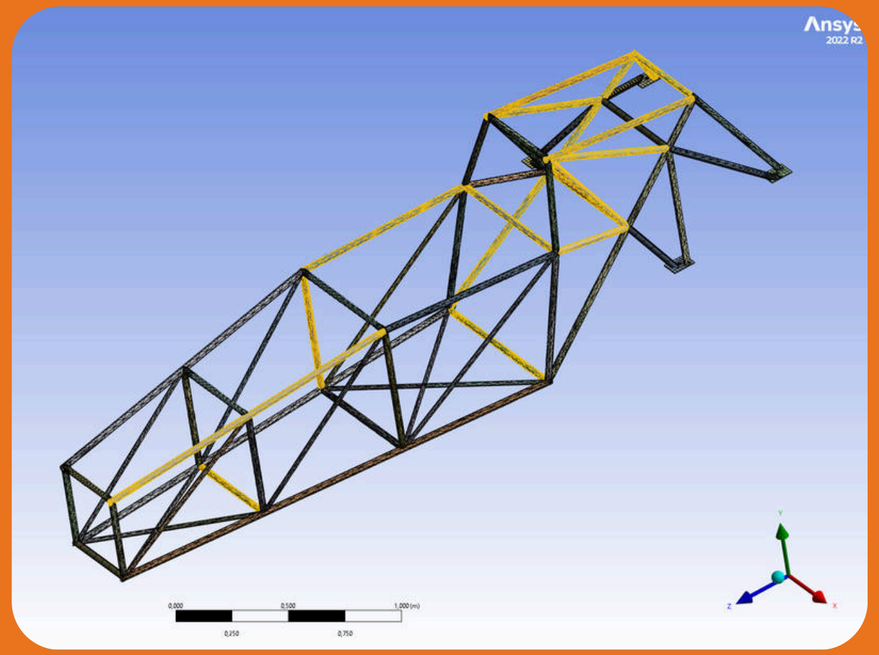
# Chassis - Assembly



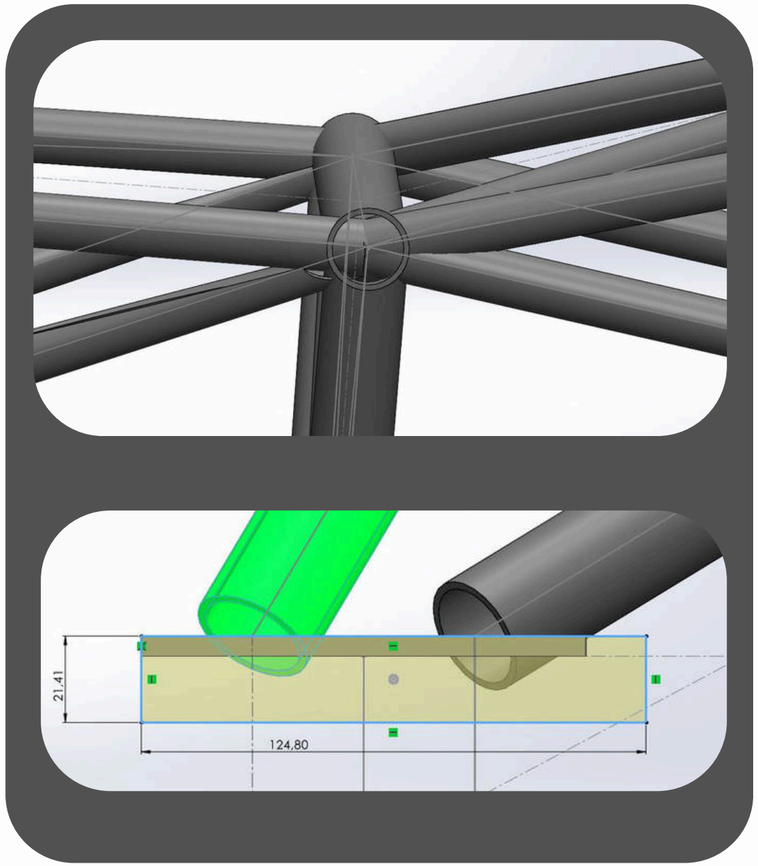

- Optimising mass and space (41/59)
- Choosing the number of batteries
- Approved by the supervisor



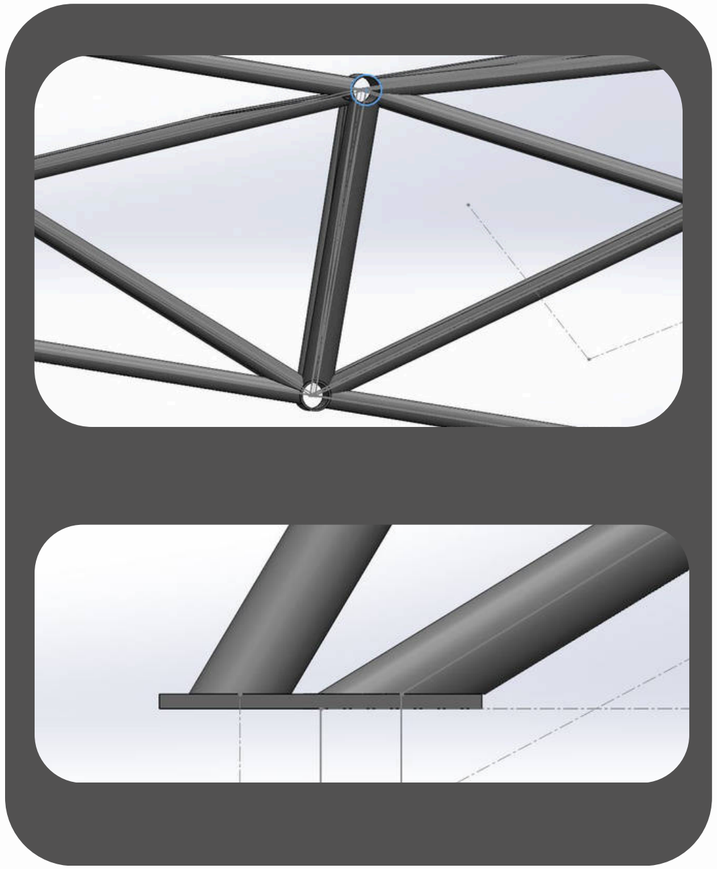
# Chassis - Structural validation




- Transfer to Ansys APDL via STEP or STP format
- Coarse mesh, hence a risk of inconsistent results
- Intersections corrected in SolidWorks




**BEFORE**



**AFTER**

# Chassis - Next steps



30CD4

- Larger tube diameter

- Checking intersections
- Structural tests

**Tests to run:**

1. At top speed
2. In a tight corner
3. Under maximum braking
4. Under maximum acceleration
5. In torsion

**Maximum limits:**

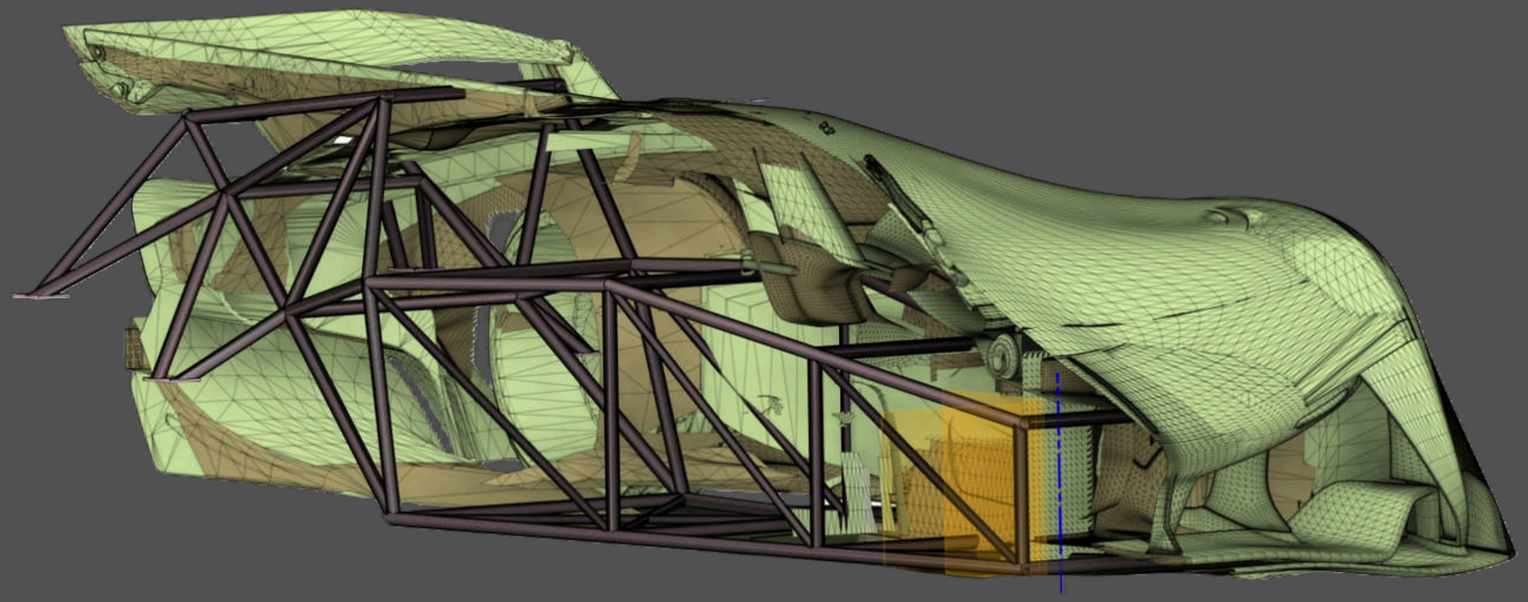
1. All stresses ≤ 460 MPa
2. All strains ≤ 0.2
3. Safety factor of 3

**Deliverables: files + reports**

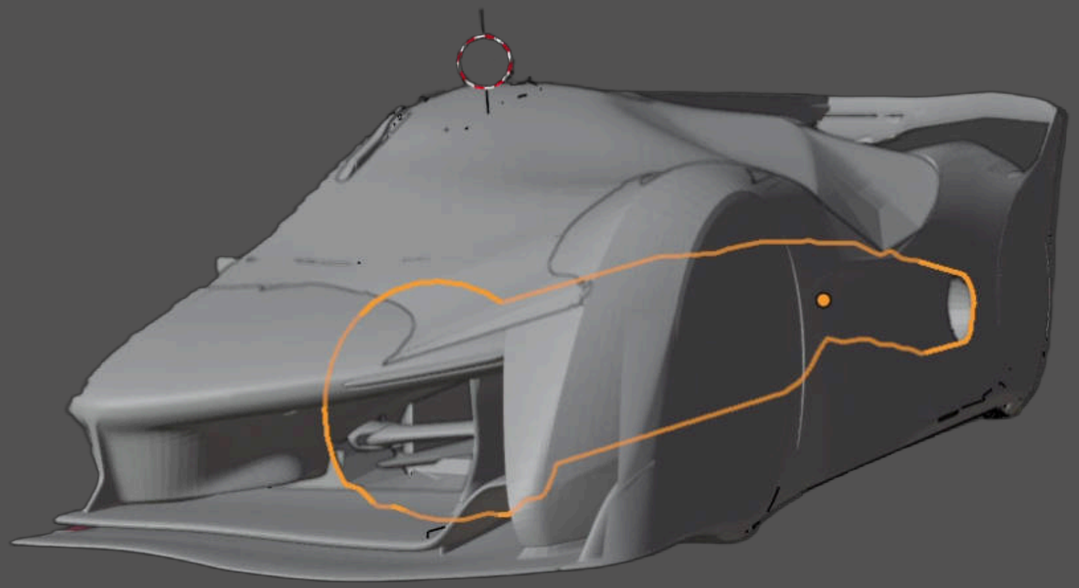
# Bodywork Study

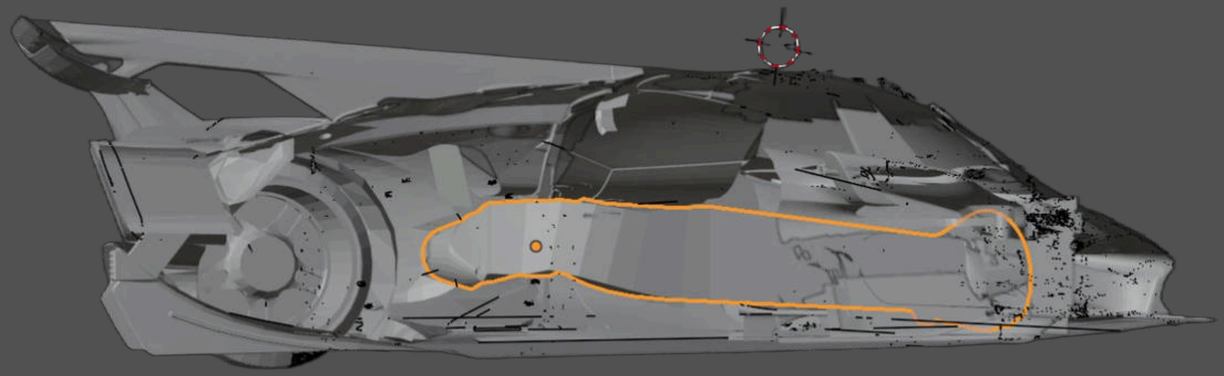
1. Make sure the bodywork fits the chassis
2. Ensure battery cooling
3. Validate the aerodynamic performance

# 1 Make sure the bodywork can accommodate the chassis



# 2 Ensure battery cooling





# 3 Validate the aerodynamic performance



ARPEX car:
Vantage

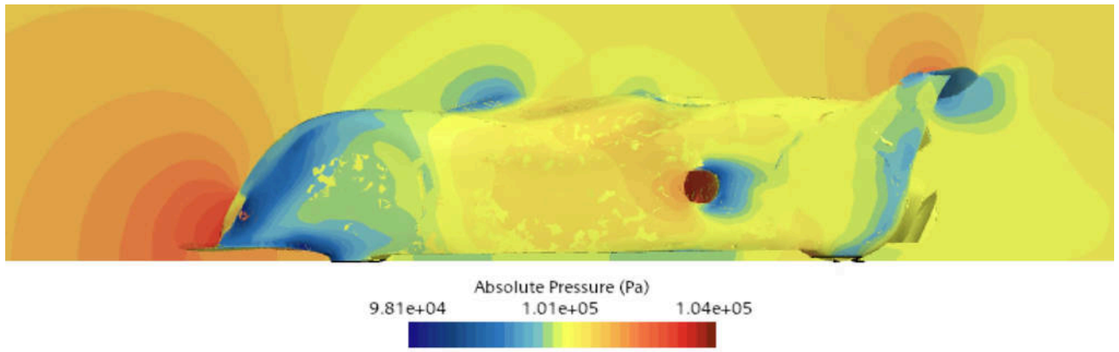


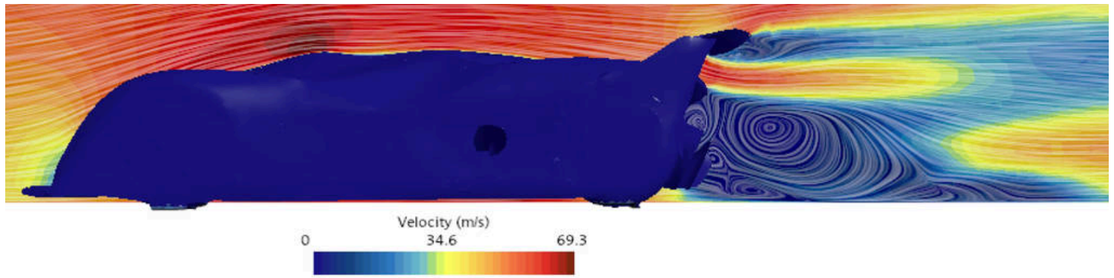

Lamborghini Vision

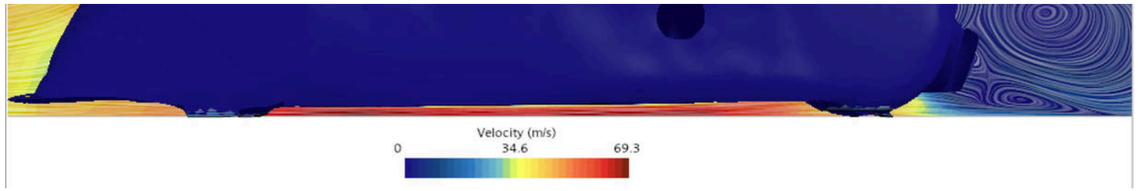
# 3 Validate the aerodynamic performance

## Qualitative study

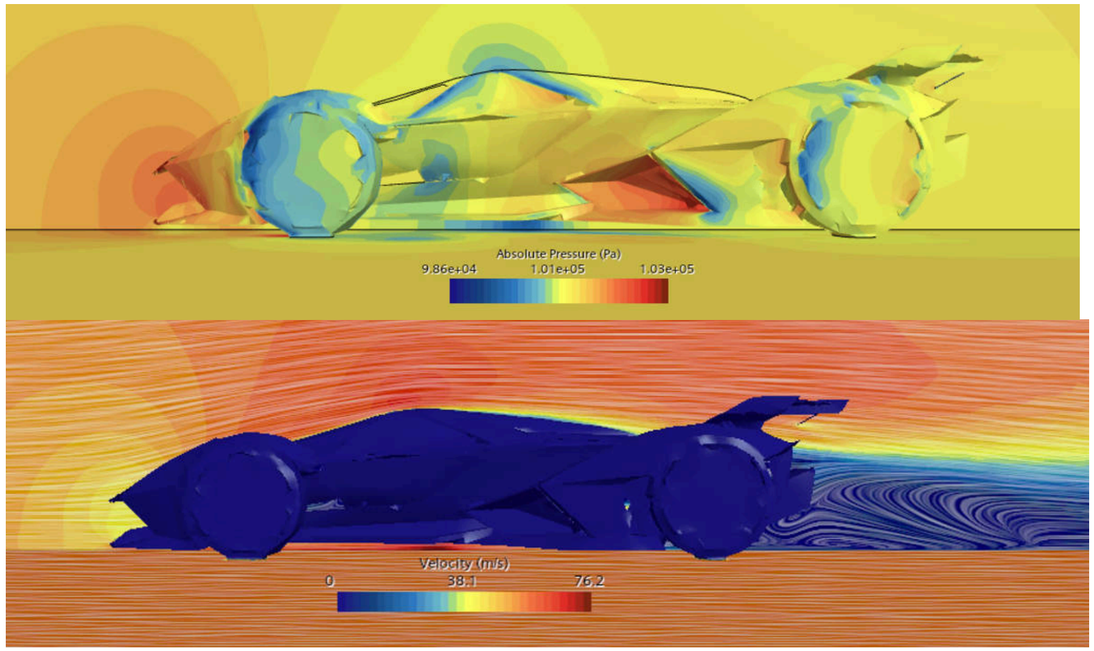
### Vantage

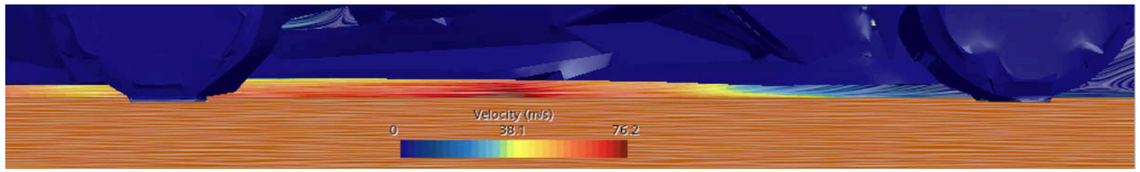



### Lamborghini Vision

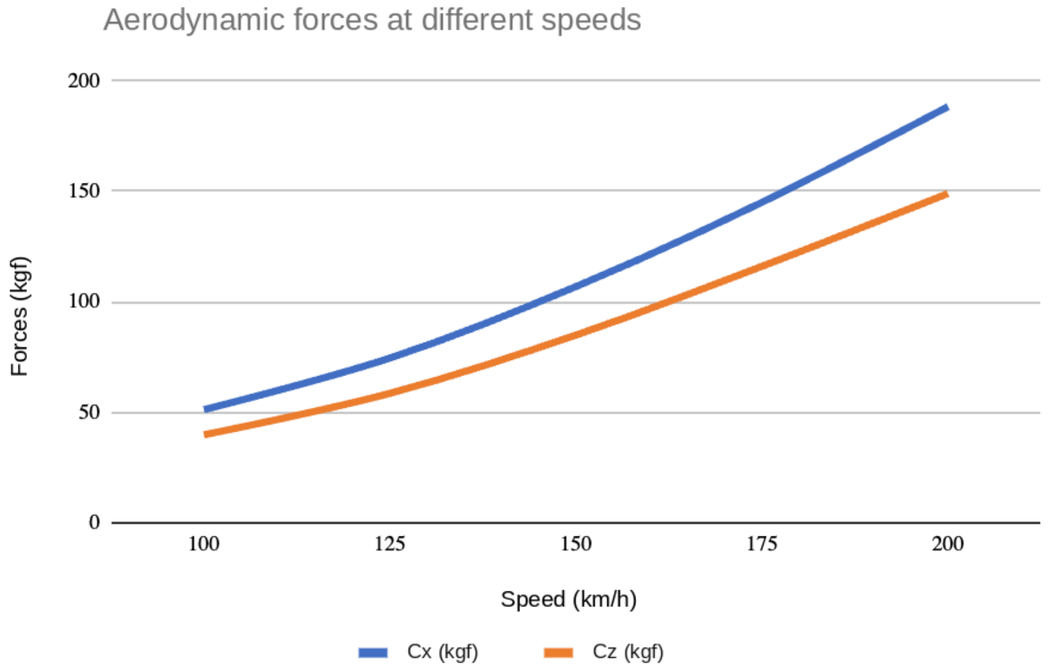


# 3 Validate the aerodynamic performance

## Quantitative study

## Vantage




200 km/h

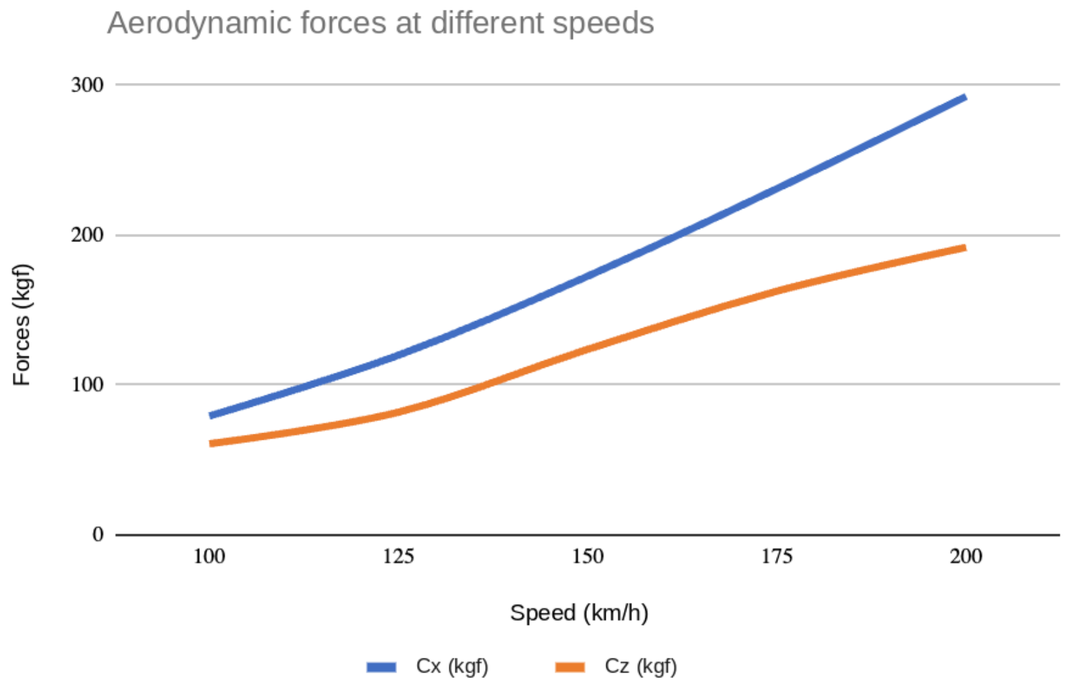
Downforce: 150 kgf

Drag force: 188 kgf

Coefficients: Cd 0.45; Cz 0.39

Cz/Cd = 0.8

## Lamborghini Vision




200 km/h

Downforce: 190 kgf

Drag force: 292 kgf

Coefficients: Cd 1.05; Cz 0.7

Cz/Cd = 0.7

# Conclusion - Aerodynamics

- ✓ Styling approved by the supervisors
- ✓ Bodywork correctly sized
- ✓ Bodywork meets the specifications
- ✗ Aerodynamic performance

## Next steps

Rework the flat floor and diffuser

Adjust the wings

Optimise the front air intakes

## Deliverables

- ✓ Simulation files
- ✓ Virtual mock-ups

# Design of an Embedded F1-Style Steering Wheel for an Electric Vehicle



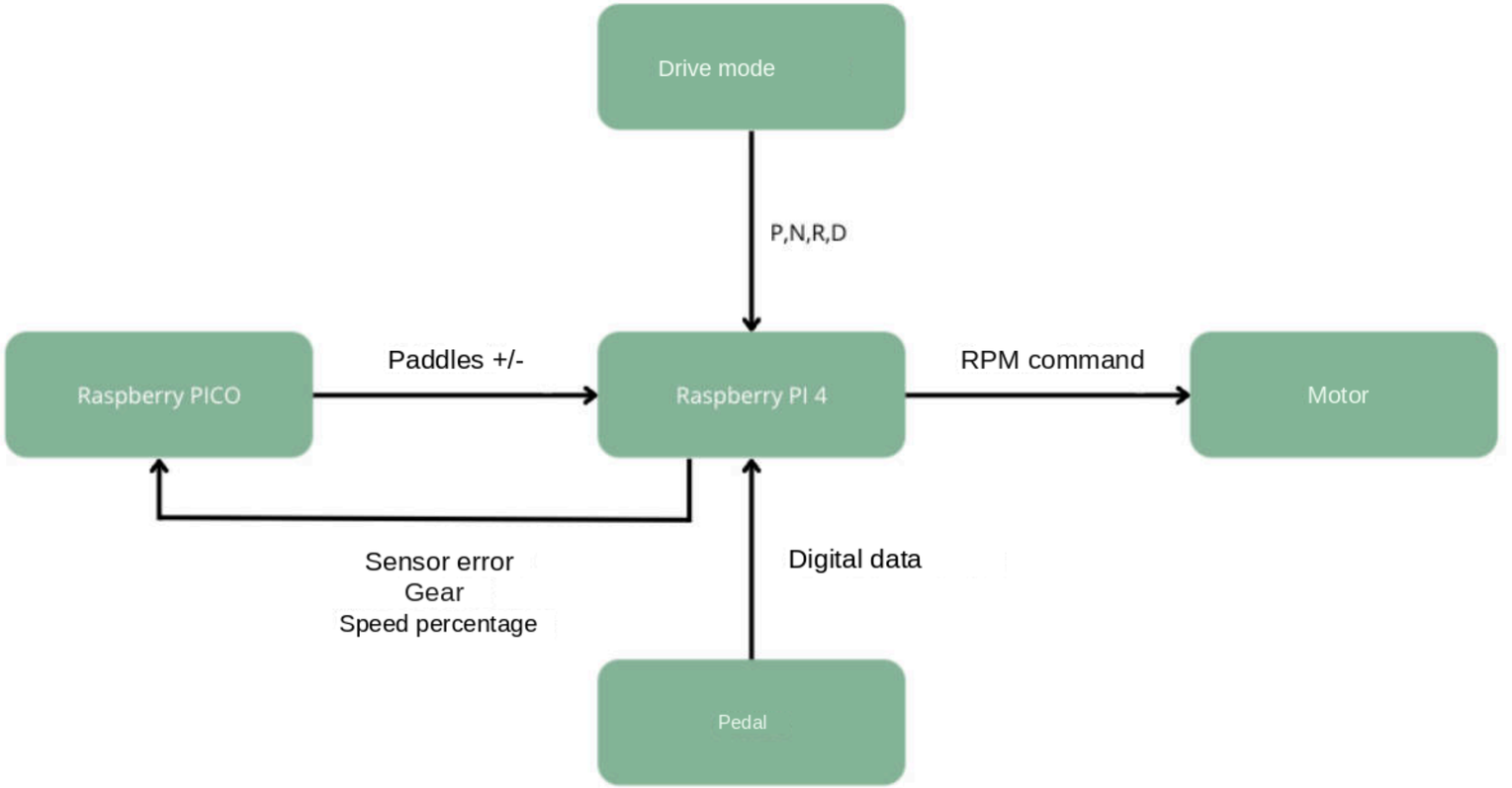
# Challenges

Constraints:

- No physical gearbox.
- Interactive interface with smooth real-time display.

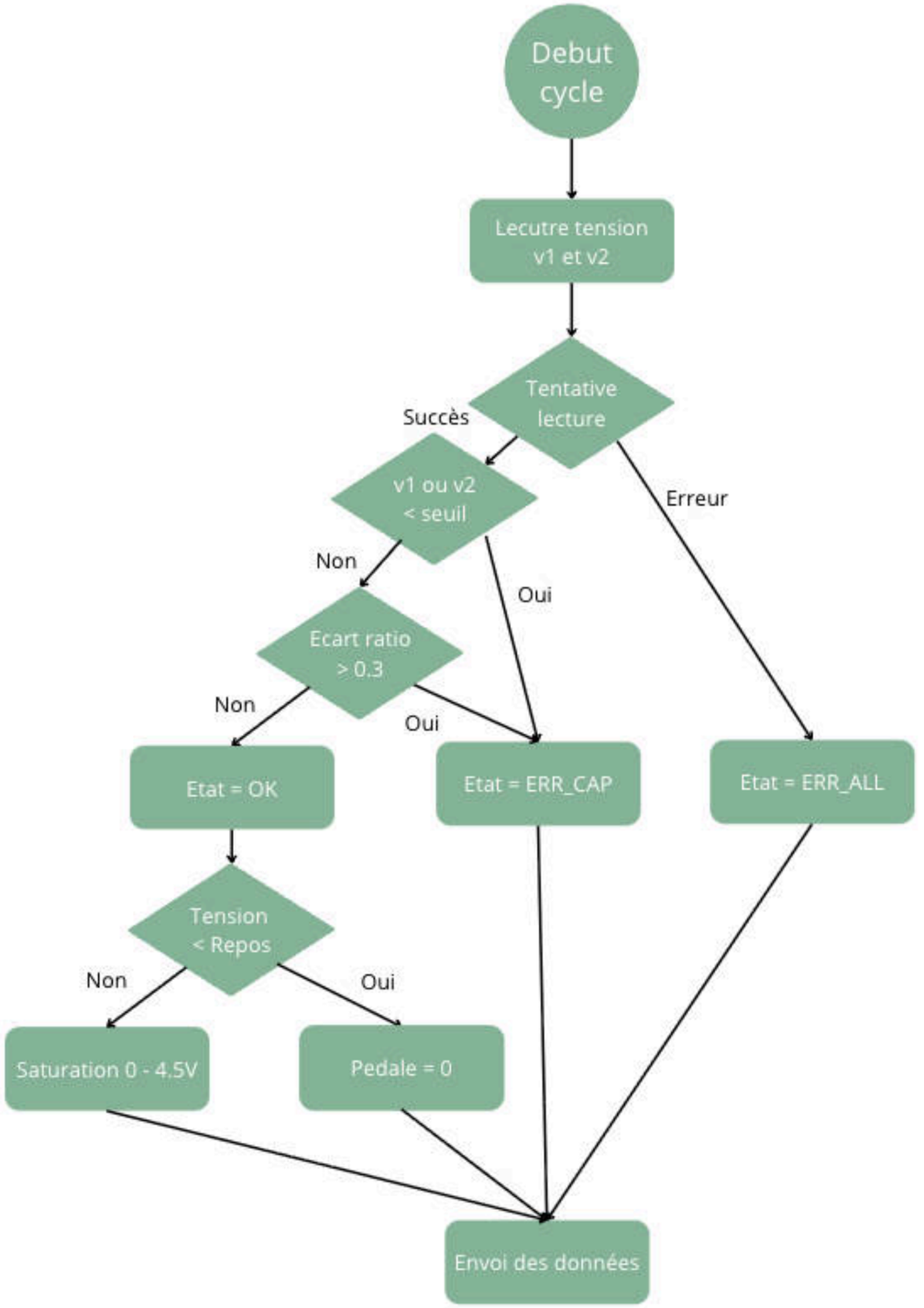
Mission:

- Centralise all critical controls in a single system and recreate, in software, the inertial behaviour of a traditional combustion-engine transmission.



# Sensor acquisition

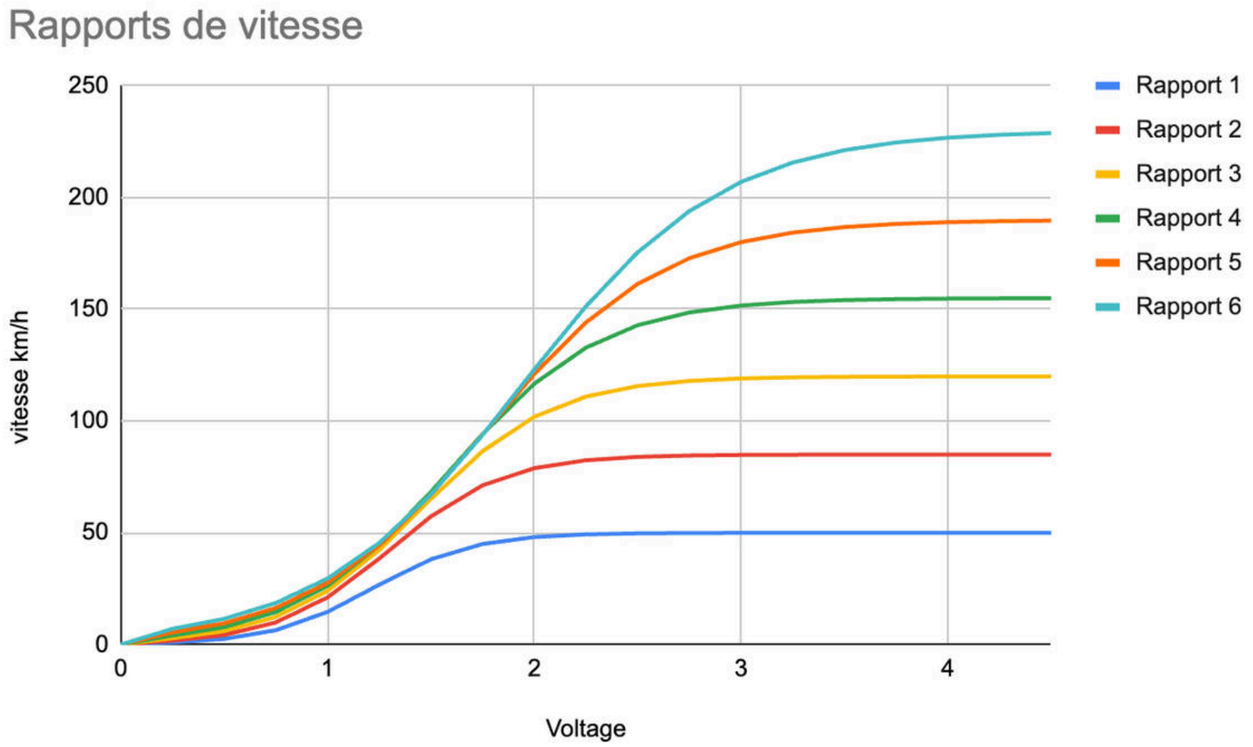
- Redundancy
- Fail-safe logic
- Protection



# Modelling the "Virtual Gearbox"

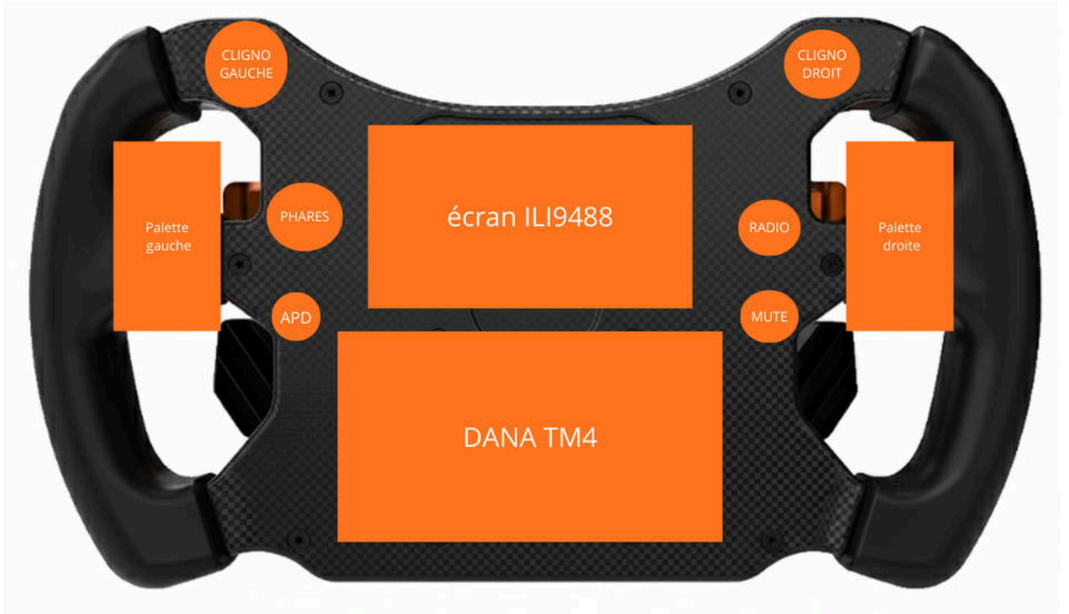
Goal: simulate a sharp acceleration followed by saturation, typical of a combustion engine

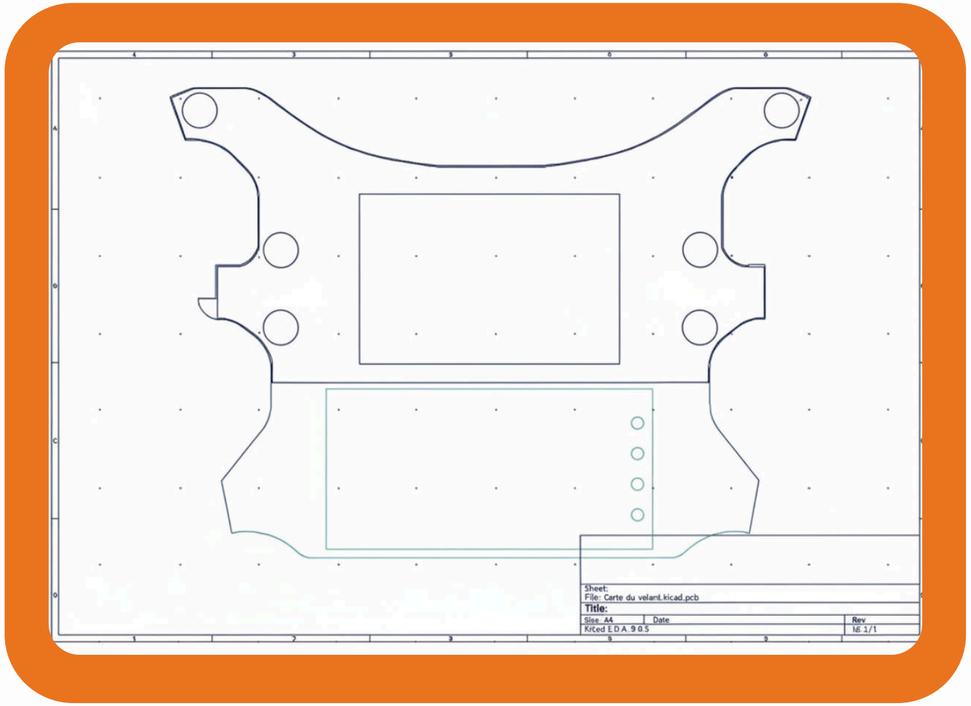
- Sigmoid function
- Equation: $$RPM_{eff} = \frac{RPM_{max\ Rapport}}{1+e^{-k(V-V_0)}}$$



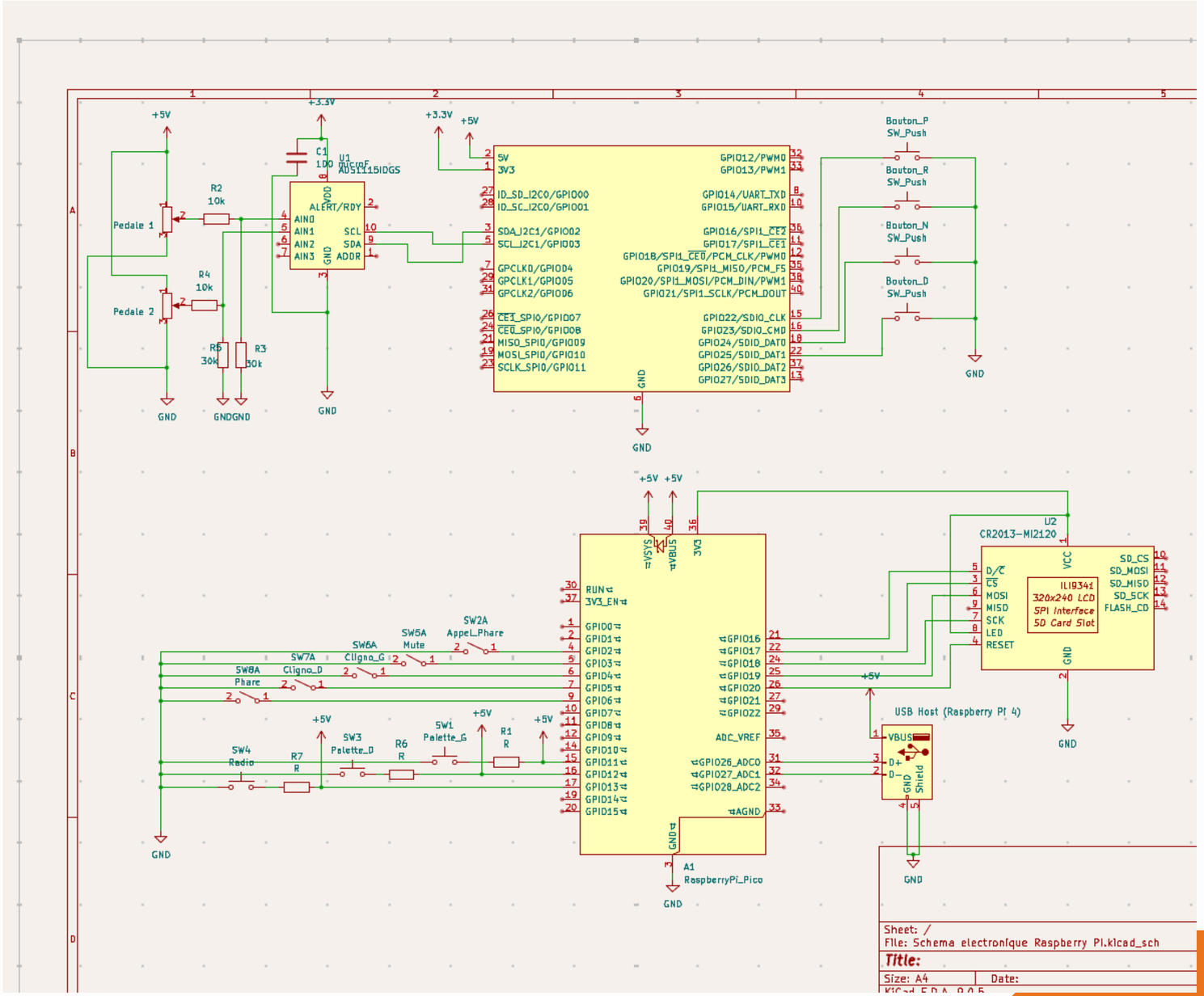
# Steering Wheel Integration

- Dual display: TFT ILI9488 + DANA TM4
- Custom PCB: routed in KiCad from an Onshape CAD extraction for a millimetre-accurate fit.

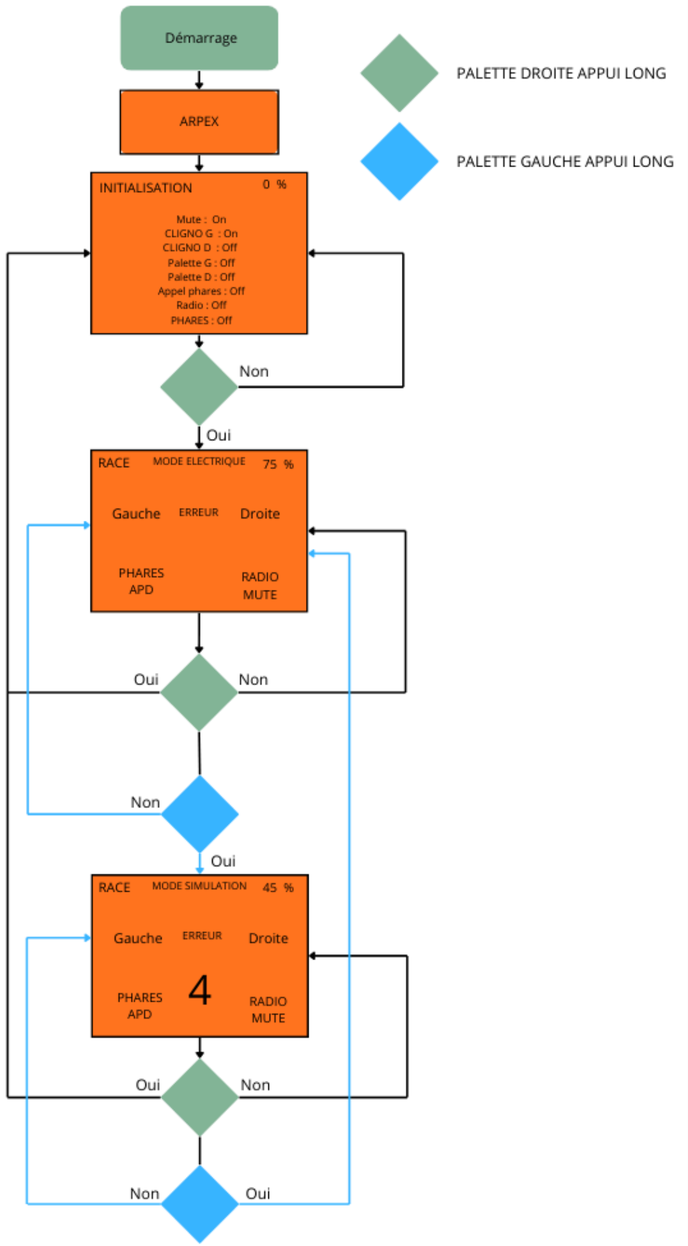


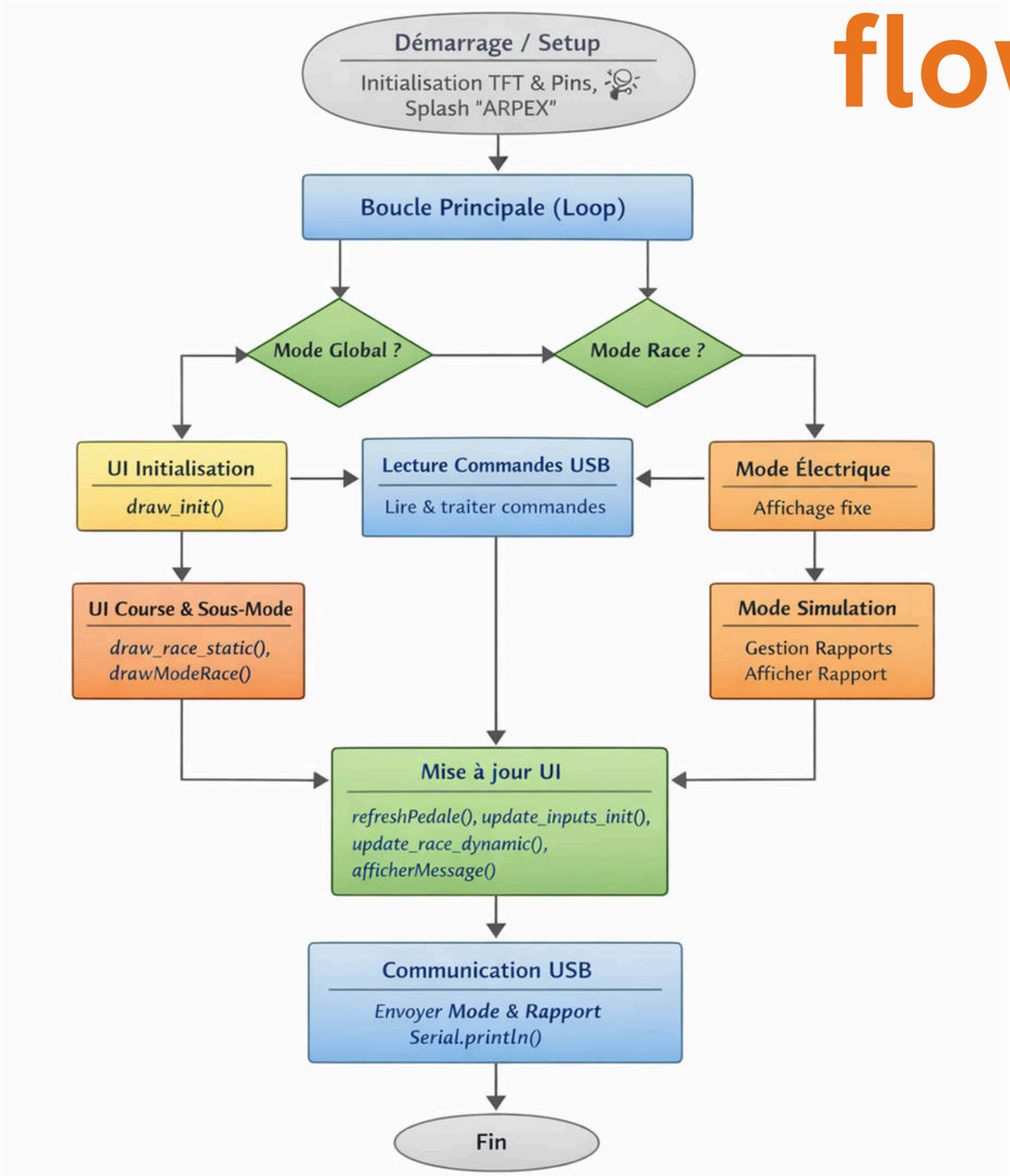
# Electronic schematic



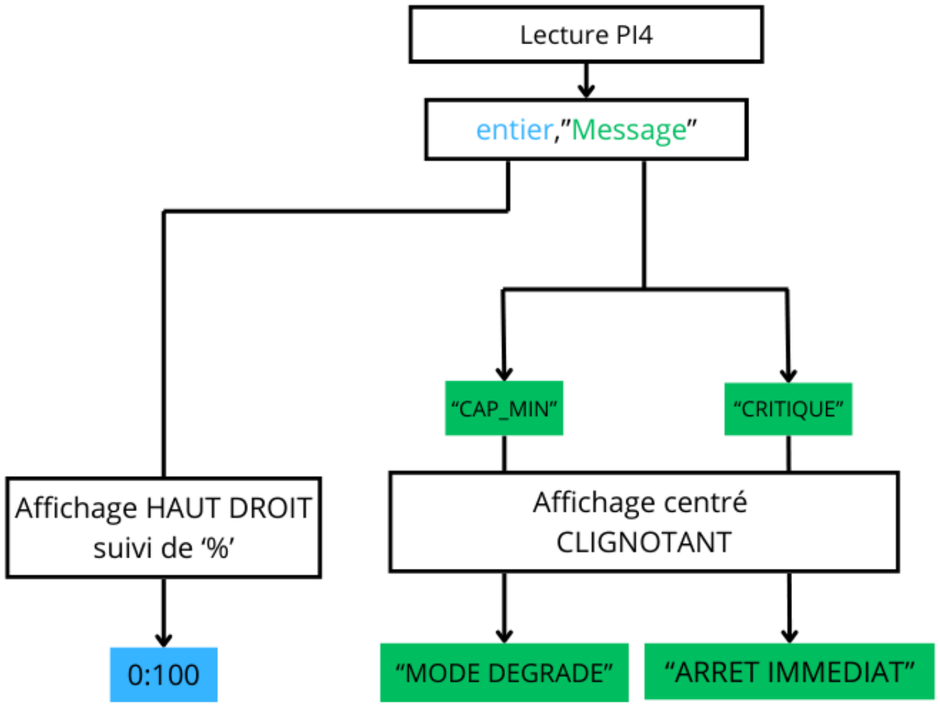
# Flow diagram

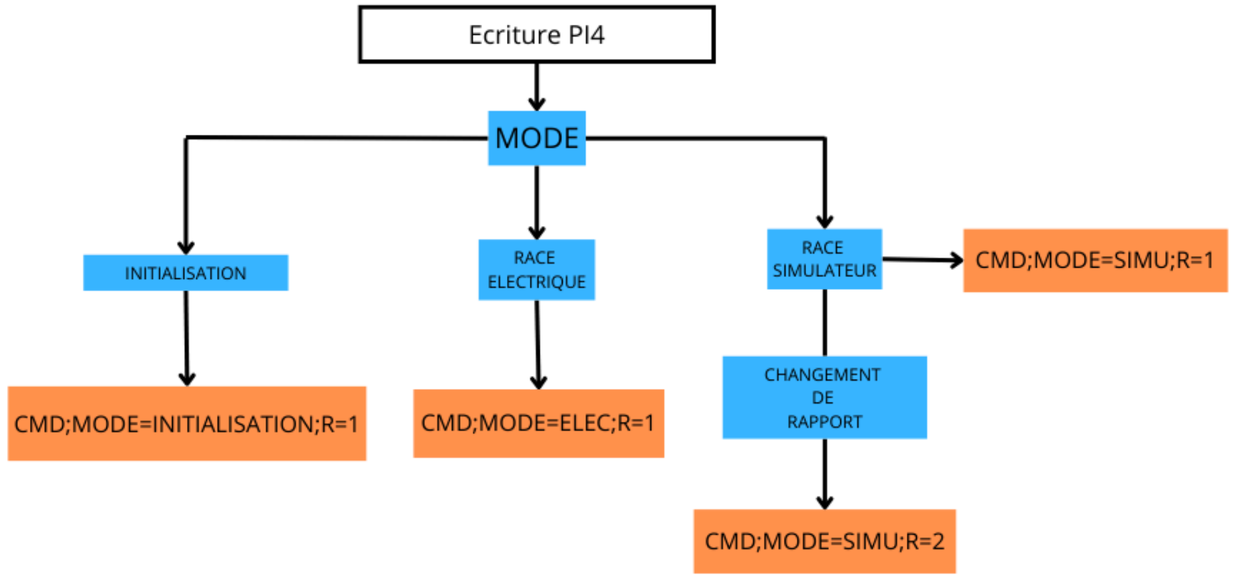


# Functional flowchart



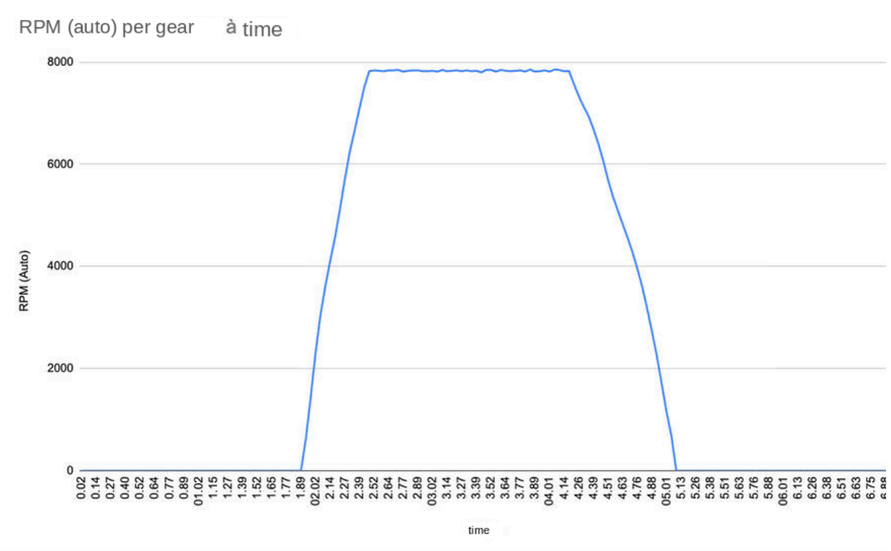
# Two-way communication PI/PICO

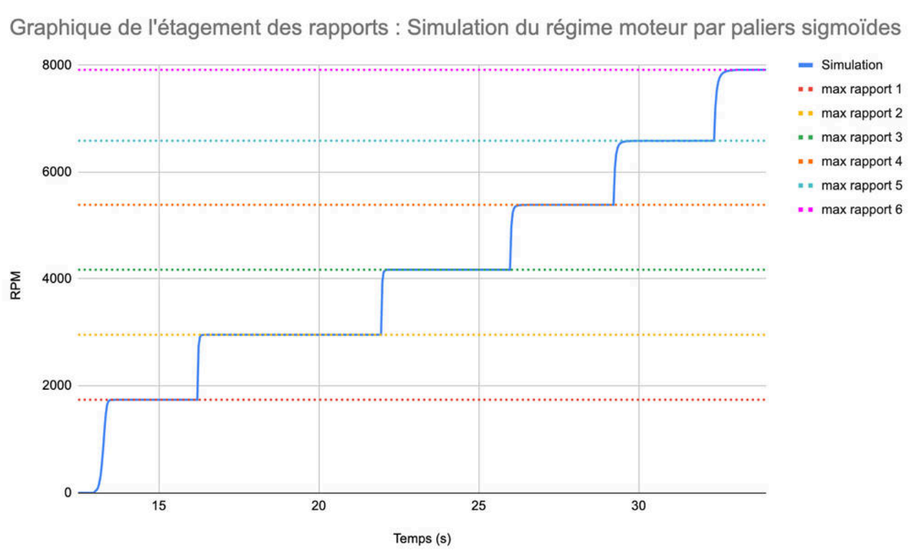


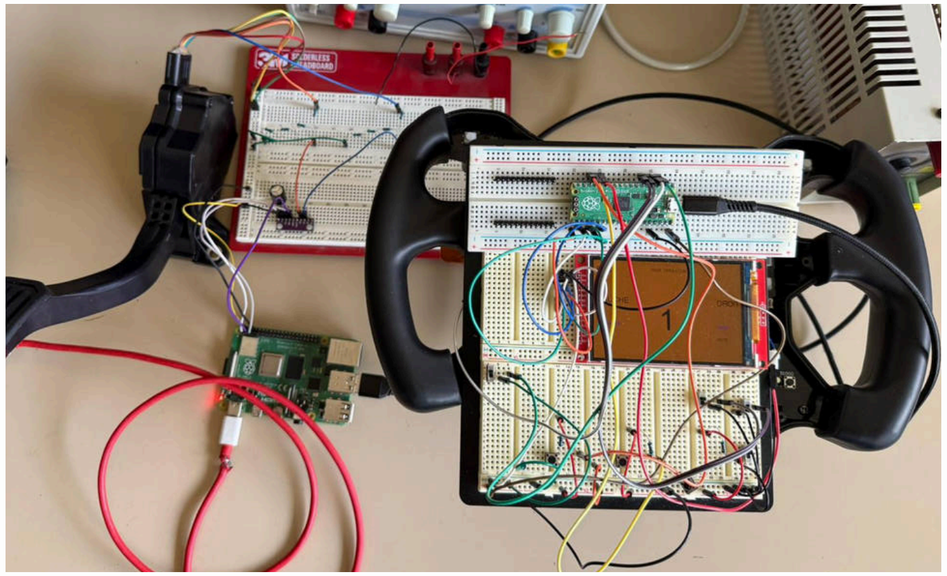
# Results

## Theoretical:




## Experimental:



# Conclusion and Outlook

- Summary:
  - Functional, safe interface integrated into the vehicle project.
  - Deliverables that let the next team carry on the work

- Outlook:
  - Add new functions
  - Improve the graphics
  - Move to a CAN bus (standard automotive multiplexing).
  - Add haptic feedback to the steering wheel to feel the motor vibrations.

# Thank you for your attention!




Any questions?

# Risk management

| Probabilité | Gravité | | | | |
|---|---|---|---|---|---|
| | 1 - Insignifiant | 2 - Mineur | 3 - Significatif | 4 - Majeur | 5 - Sévère |
| 5 - Très probable | Absences | | Contraintes budgétaires | | |
| 4 - Probable | | | Compétences logicielles => Apprentissage continu | Manque de temps => Préparer une bonne continuité | |
| 3 - Possible | | Manque de données | Évolution normative | Perte de données des simulations => Rassembler sur drive | |
| 2 - Peu probable | | | Coordination des équipes | Manque de motivation | |
| 1 - Très peu probable | | | | | |

| |
|---|
| Peut compromettre le projet |
| À Surveiller |
| Risque mineur |
| Insignifiant |